

# 12: Linux MMIO

Engr 315: Hardware / Software Codesign

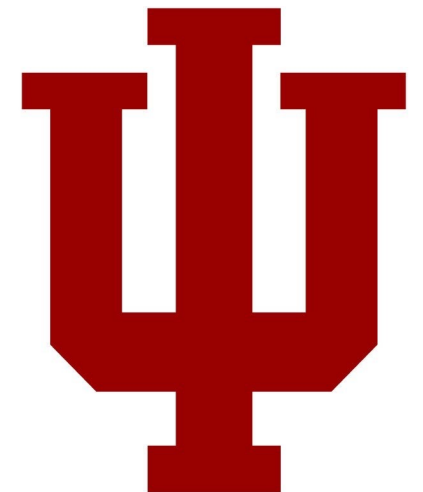
Andrew Lukefahr

*Indiana University*

Some material taken from:

EECS 373 & EECS 370 University of Michigan

<https://developer.arm.com/documentation/102202/0300/Transfer-behavior-and-transaction-ordering>

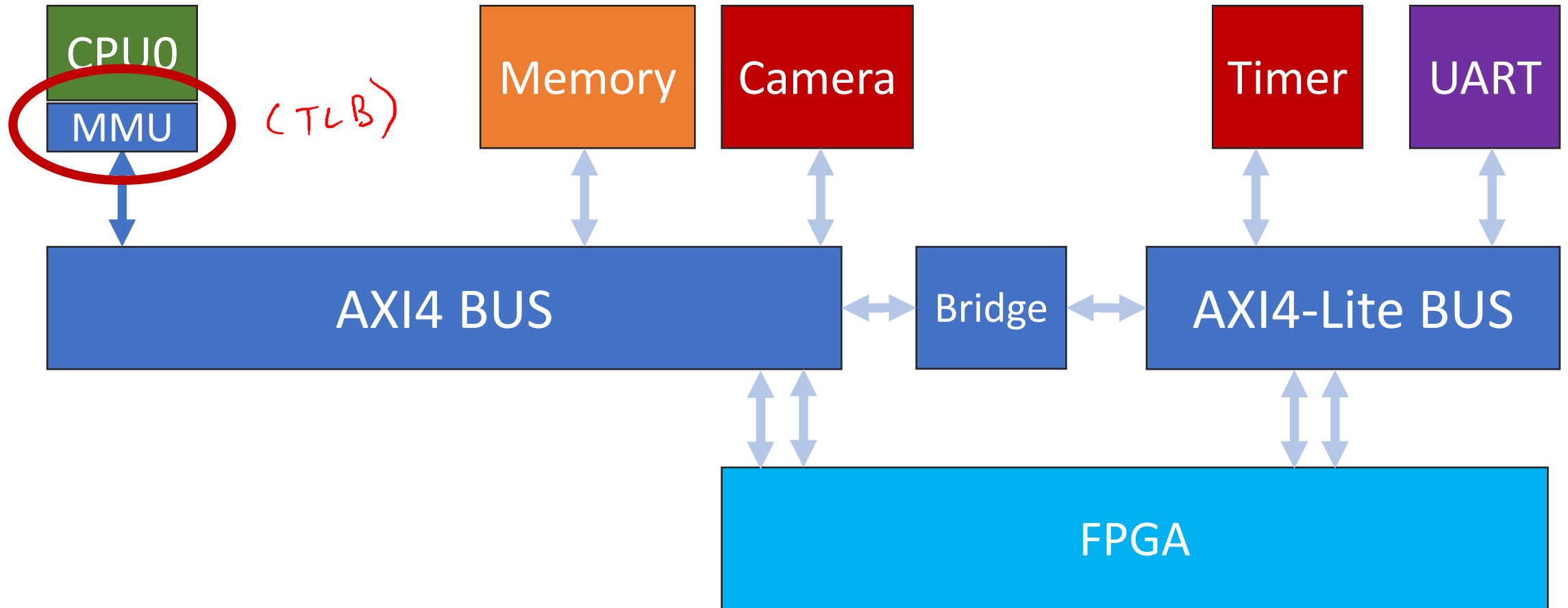


# Exam Planning

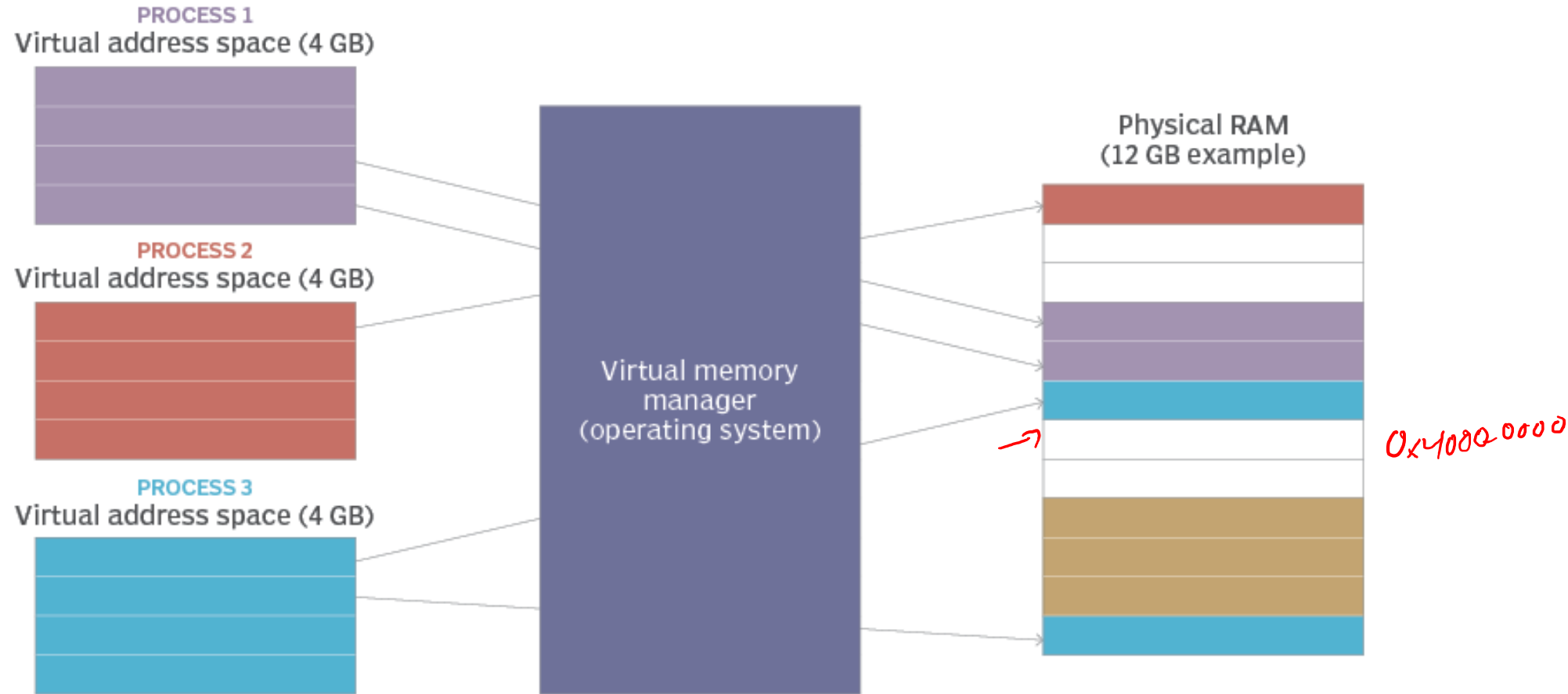
|       |           |    |        |   |
|-------|-----------|----|--------|---|
| 11/01 | Wednesday | 18 | Review | – |
| 11/06 | Monday    | 19 | Exam   |   |
| 11/08 | Wednesday | 20 | Review |   |

<https://engr315.github.io>

# Machine Model, V3: MMUs



# OS (Linux) mains full Virtual->Physical Mappings

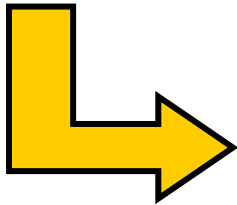


# TLB Address Translation

Valid Translation Look-Aside Buffer (TLB)

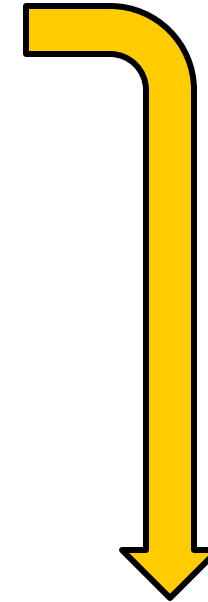
Virtual Address  
0x4000-0000

| Virtual Base | Offset |
|--------------|--------|
|--------------|--------|



0x2000-0000

| v | Virtual Base | Physical Base |
|---|--------------|---------------|
|   | 4000         | 5000          |
|   | 3000         | 3000          |
|   | 2000         | 4000          |
|   |              |               |
|   |              |               |
|   |              |               |
|   |              |               |

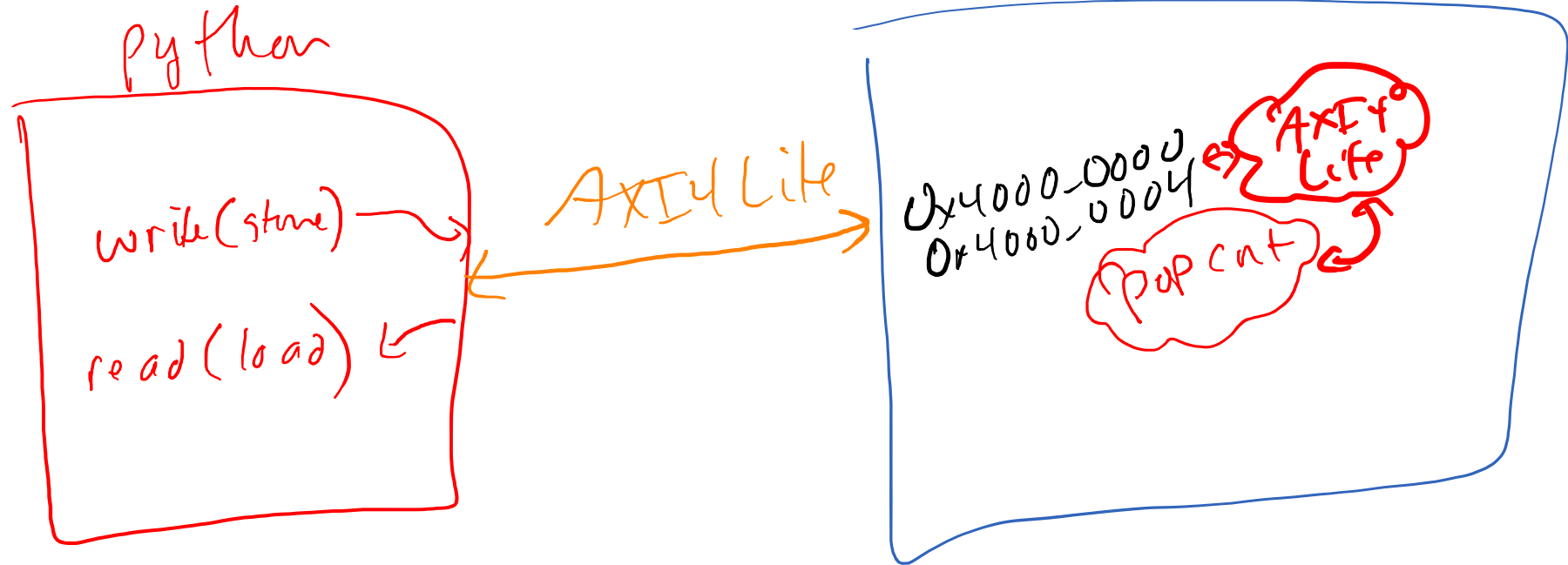


| Physical Base | Offset |
|---------------|--------|
| 5000          | 0000   |

Physical Address  
4000-0000

# P4: MMIO Popcount

FPGA



# Vivado MMIO EMA

```
module ema(  
    input          ACLK,  
    input          ARESETN,  
    input [31:0]    WRITE_DATA,  
    input          WRITE_VALID,  
    output logic [31:0] READ_DATA,  
    input          READ_VALID  
);  
    logic [31:0] y_last;  
    logic [31:0] y_hist;  
    assign READ_DATA = (y_last >> 2) +  
        (y_hist>> 1) + (y_hist >> 2);  
  
    always@(posedge ACLK) begin  
        if (~ARESETN) begin  
            y_last <= 32'h0;  
            y_hist <= 32'h0;  
        end else if (WRITE_VALID)  
        begin  
            y_last <= WRITE_DATA;  
            y_hist <= READ_DATA;  
        end  
    end  
endmodule
```

# Vivado MMIO EMA

```
ema ema0(  
    .ACLK(S_AXI_LITE_ACLK) ,  
    .ARESETN(S_AXI_LITE_ARESETN) ,  
  
    .WRITE_DATA(WRITE_MEM[0]) ,  
    .WRITE_VALID(WRITE_MEM_VALID[0]) ,  
  
    .READ_DATA(READ_MEM[0]) ,  
    .READ_VALID(READ_MEM_VALID[0])  
);
```

*high for 1 cycle to indicate new data*

*high for 1 cycle to indicate read.*

Question:

- How do we access the FPGA from C w/Linux?
- FPGA uses **physical** address
- CPU (w/Linux) uses **virtual** address

Answer:

- Linux lets us cheat.

# Python Example

```
from pynq import Overlay
from pynq import MMIO
class hw_ema():
    def __init__(self):
        self.overlay = Overlay('bitstream.bit')
        self.mmio = self.overlay.axi_popcount_0.S_AXI_LITE
    def ema(self, n):
        self.mmio.write(0x0, int(n))
        return self.mmio.read(0x0)
ema = hw_ema()
for i in range(1000, 6000, 1000):
    x = ema.ema(i)
    print ("In: ", i, " Out: ", x)
```

```
$ sudo python3 mmio_demo.py
```

```
In:  1000   Out:  250
```

```
In:  2000   Out:  687
```

```
In:  3000   Out: 1264
```

```
In:  4000   Out: 1948
```

```
In:  5000   Out: 2711
```

# /dev/mem

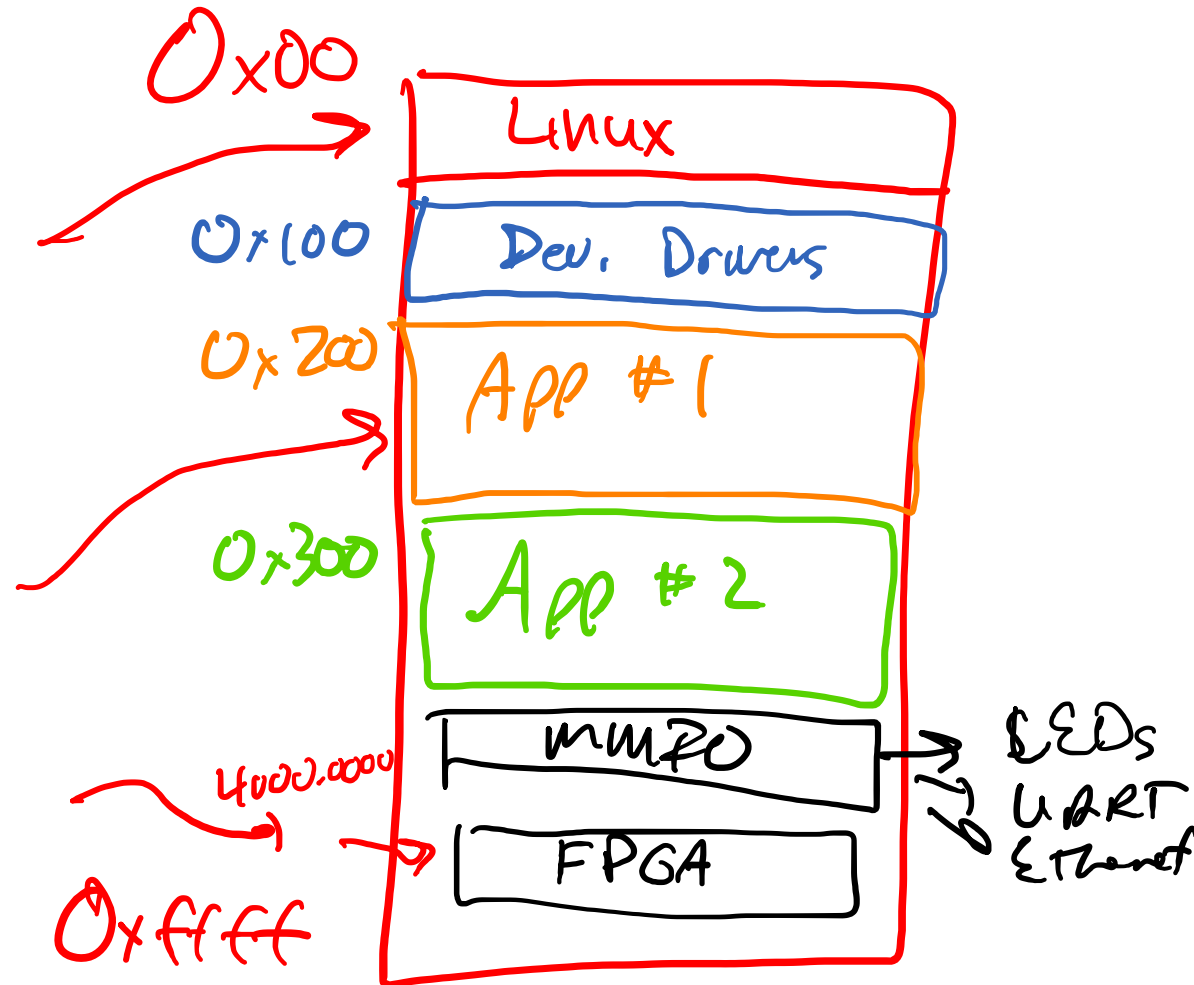
- */dev/mem* is a character device file that **is an image of the main memory** of the computer. It may be used, for example, to examine (and even patch) the system.
- Byte **addresses in */dev/mem* are interpreted as physical memory addresses**. References to nonexistent locations cause errors to be returned.
- Requires root (sudo) access



/dev/mem

really powerful  
→ really dangerous

→ Pyng does  
this?!?!?

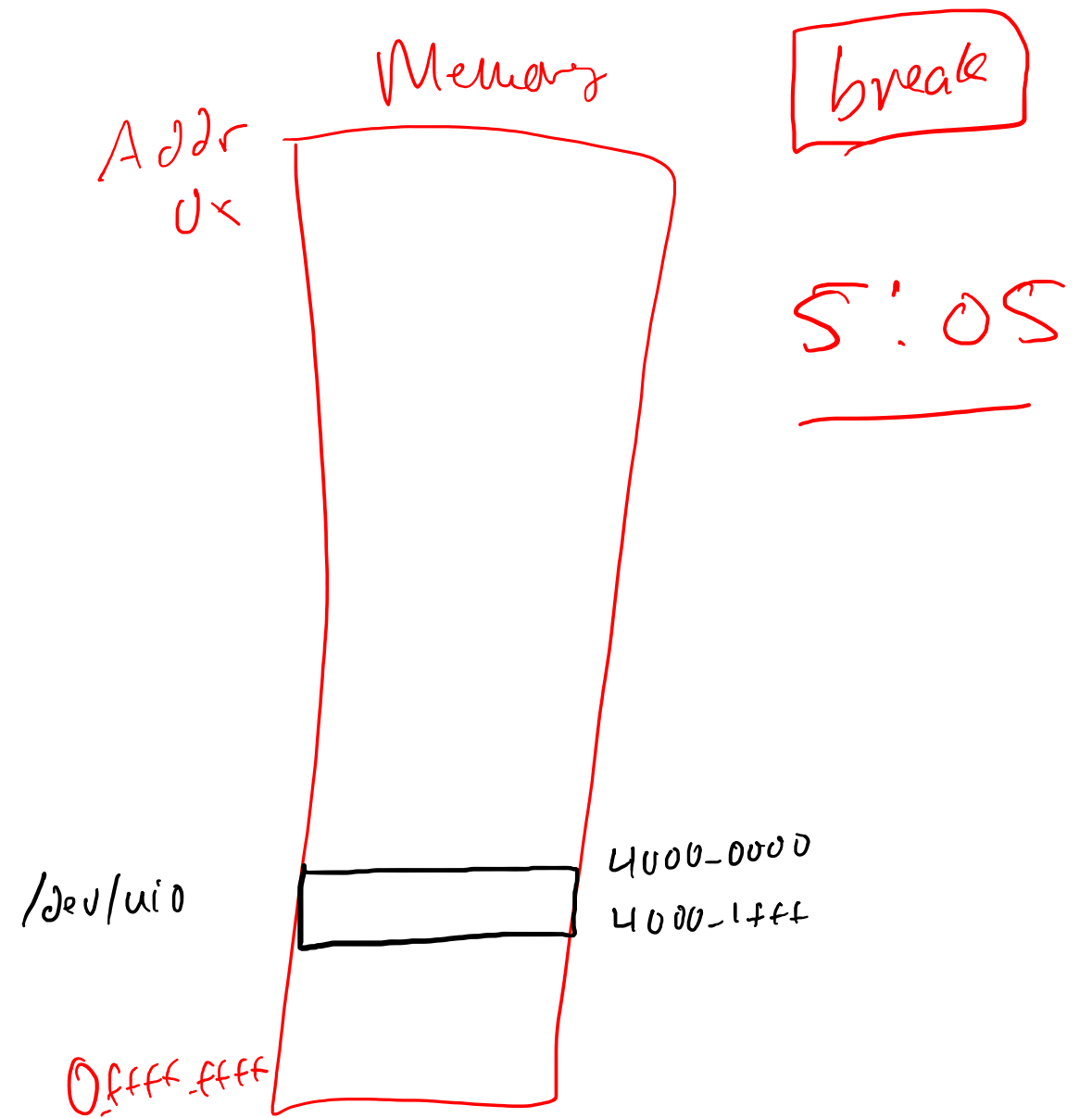


# Userspace I/O

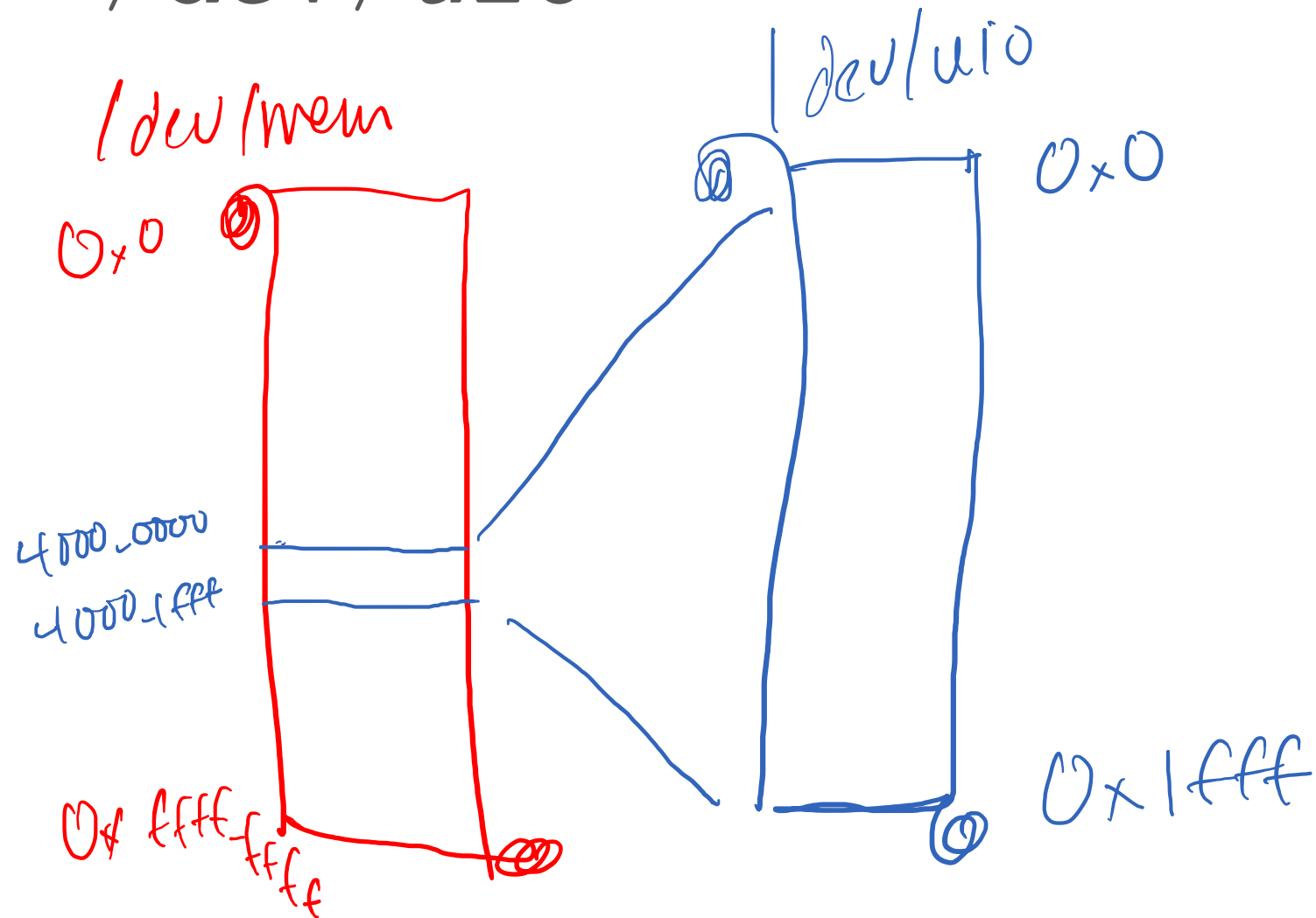
- Linux provides a more restrictive version of /dev/mem called /dev/uio
- Linux exposes a small portion of I/O memory space to the user through /dev/uioX interfaces.
- Treat it just like /dev/mem, but not the entire memory space

User Accessible 😊

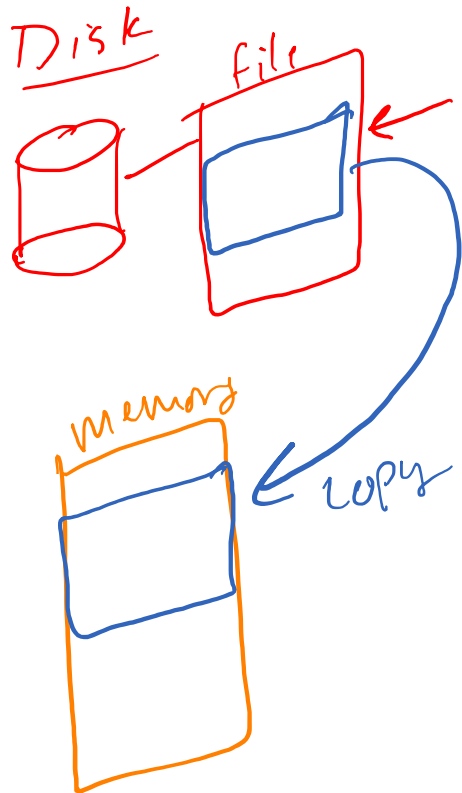
/dev/uio



/dev/uio



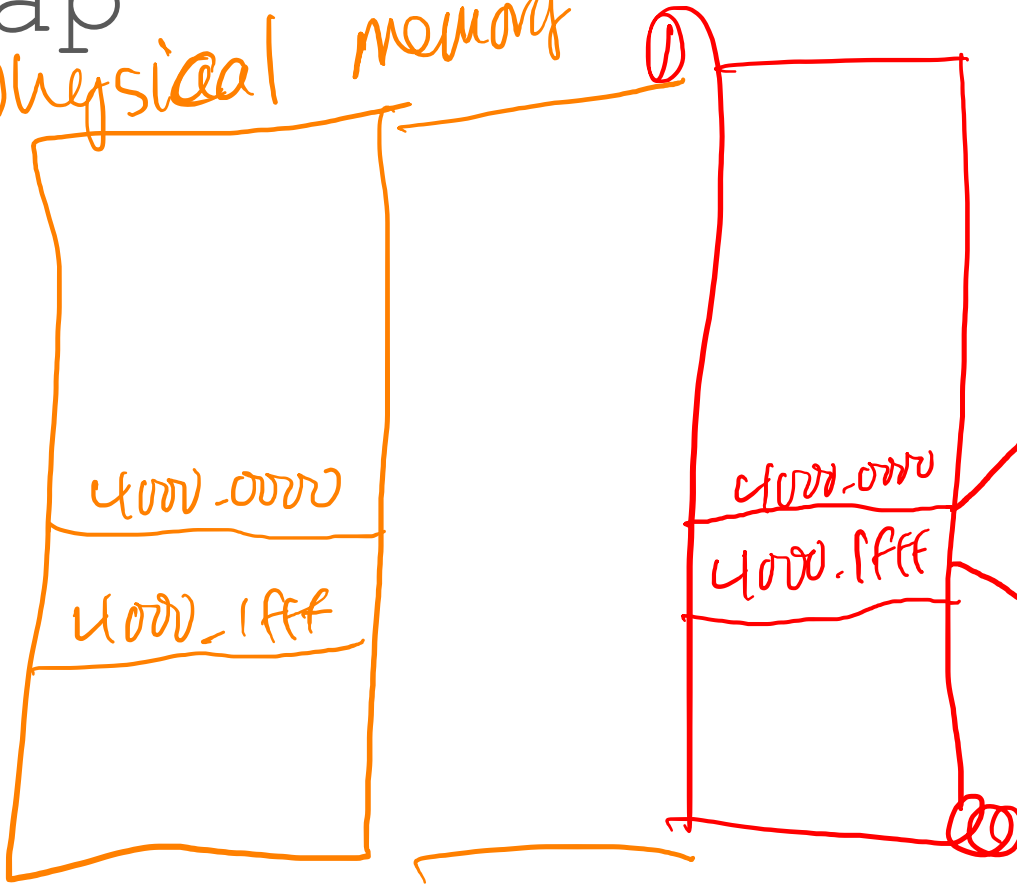
# mmap – memory map



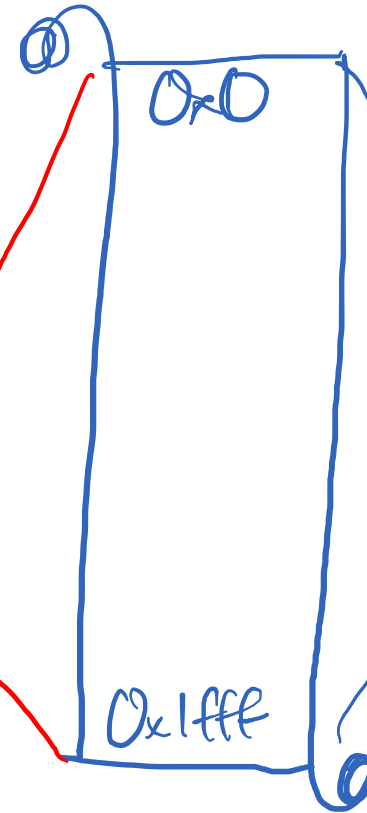
- **mmap()** creates a new mapping in the virtual address space of the calling process. The starting address for the new mapping is specified in *addr*.
- The contents of a file mapping are initialized using *length* bytes starting at offset *offset* in the file (or other object) referred to by the file descriptor *fd*.

mmap

physical memory

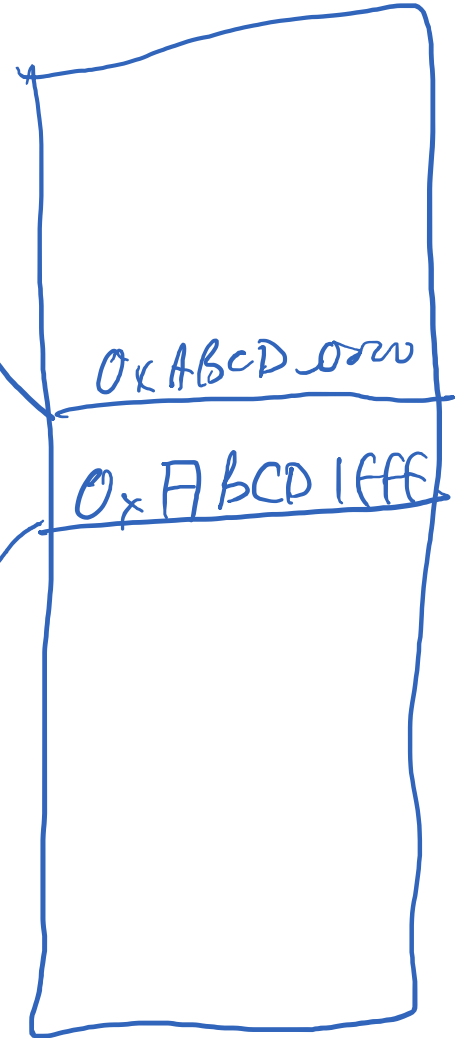


Dev/mem



/dev/mem

App Mem



# Userspace EMA w/C

```
int main (){

    int dev_mem_fd = -1;
    void * vaddr_base;

    //Mapping user-space I/O
    dev_mem_fd = open("/dev/uio0", O_RDWR|O_SYNC);
    if (dev_mem_fd < 0) { perror("open() /dev/uio0"); return 1; }

    // Map 1KB of physical memory starting at uio 0x0 (real 0x40000000)
    // to 1KB of virtual memory starting at vaddr_base
    vaddr_base = mmap(0, 1024, PROT_READ|PROT_WRITE,
        MAP_SHARED, dev_mem_fd, 0x0);

    if (vaddr_base == MAP_FAILED) { perror("mmap()"); return 1; }
```

```
volatile uint32_t * ema_reg = (uint32_t*) vaddr_base;
uint32_t tmp;

for (int i = 1000; i < 6000; i +=1000){
    printf ("Sending in: %d\n", i);
    *ema_reg = i; //mmio store
    tmp = *ema_reg; //mmio load
    printf("Receiving: %d\n", tmp);
}

if (munmap(vaddr_base, 1024) != 0) { perror("munmap()"); }
if (close(dev_mem_fd) != 0) { perror("close()"); }
dev_mem_fd = -1;

return 0;
}
```

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uint32_t tmp;

for (int i = 1000; i < 6000; i +=1000){
    printf ("Sending in: %d\n", i);
    *ema_reg = i; //mmio store to 0x4000_0000
    tmp = *ema_reg; //mmio load from 0x4000_0000
    printf("Receiving: %d\n", tmp);
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```

# Complete EMA

```
#include <fcntl.h>
#include <stdlib.h>
#include <stdint.h>
#include <stdio.h>
#include <sys/mman.h>
#include <unistd.h>

int main (){

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if (munmap(vaddr_base, 1024) != 0) { perror("munmap()"); }
if (close(dev_mem_fd) != 0) { perror("close()"); }
dev_mem_fd = -1;

return 0;
}
```

# Demo Time

# Next Time:

- DMA

# 10: Linux MMIO

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